DOHYEON (DANIELLE) KIM

Applied and Computational Mathematics, California Institute of Technology
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RESEARCH INTERESTS

Interacting particle systems and mean-field limits; propagation of chaos, long-time asymptotics, fluctuations around the mean-field limit, and large deviation principles; stochastic differential equations and kinetic equations; mathematical analysis of particle methods for sampling and nonlinear filtering; classical and quantum dynamics for optimization.

EDUCATION

California Institute of Technology, USA *Sep 2021 – Expected Jun 2027*
Ph.D. Candidate in Applied and Computational Mathematics GPA: 4.24/4.3
Advisor: Prof. Franca Hoffmann (since May 2023)

Yonsei University, South Korea
BBA in Business Administration and BS/MS in Mathematics GPA: 4.29/4.3 (4.0/4.0)
Graduated with Highest Distinction; Valedictorian (1/2674)
Master's thesis: "Solvability of Second-Order Elliptic and Parabolic Equations with VMO-Type Coefficients"
Advisor: Prof. Seick Kim

Columbia University, USA
One-year undergraduate study abroad program GPA: 3.9/4.0
Completed eight upper-level mathematics courses

PUBLICATIONS AND PREPRINTS

Authors are listed alphabetically, as is customary in mathematics.

From Consensus-Based Optimization to Particle Swarm Optimization: Convergence Guarantees under Drift-Diffusion Coupling
Franca Hoffmann, **Dohyeon Kim**, Ritvik Teegavarapu
Accepted for poster presentation at the NeurIPS 2026 DynaFront Workshop.
arXiv:2609.17002 (September 2026)

Quantitative Target Convergence and Uniform-in-Time Propagation of Chaos for Langevin-Regularized SVGD
Sayan Banerjee, **Dohyeon Kim**
Submitted, arXiv:2608.28827 (August 2026)

Long-time Stability and Convergence of Particle Swarm Optimization
Giacomo Borghi, Hui Huang, **Dohyeon Kim**
Submitted, arXiv:2607.24696 (July 2026)

Uniform-in-time propagation of chaos for Second-Order Consensus-Based Optimization
Seung-Yeal Ha, Franca Hoffmann, **Dohyeon Kim**
Preprint, arXiv:2605.20697 (May 2026)

Uniform-in-time propagation of chaos for Consensus-Based Optimization
Nicolai Gerber, Franca Hoffmann, **Dohyeon Kim**, Urbain Vaes
Accepted for publication in *SIAM Journal on Mathematical Analysis* (October 2026).
arXiv:2505.08669 (May 2025)

MirrorCBO: A consensus-based optimization method in the spirit of mirror descent
Leon Bungert, Franca Hoffmann, **Doh Yeon Kim**, Tim Roith
Mathematical Models and Methods in Applied Sciences, 2025.
arXiv:2501.12189 (January 2025)

Mono-cluster flocking and uniform-in-time stability of the discrete Motsch–Tadmor model
Seung-Yeal Ha, Franca Hoffmann, **Dohyeon Kim**, Wook Yoon
Communications on Pure and Applied Analysis **32** (2026), 234–281.
doi:10.3934/cpaa.2026040; arXiv:2408.10213 (2024)

MANUSCRIPTS IN PREPARATION

Long-Time Behavior of Continuous-Time Consensus-Based Sampling
Dohyeon Kim
Manuscript in preparation

Refined Long-Time Asymptotics for Variants of Consensus-Based Optimization
Franca Hoffmann, **Dohyeon Kim**, Wuchen Li
Manuscript in preparation.

Mean-Field Well-Posedness and Convergence Guarantees for Localized Consensus-Based Sampling
Arne Bouillon, Kathrin Hellmuth, Franca Hoffmann, **Dohyeon Kim**
Manuscript in preparation.

CONFERENCES, TALKS, AND RESEARCH VISITS

Oberwolfach Workshop: Variational Methods for Evolution *Dec 2026*
Invited participant
Mathematisches Forschungsinstitut Oberwolfach, Germany

Research Visit at BICMR *Oct 17–25, 2026*
Hosted by Prof. Zhenfu Wang
Talk in Prof. Zhenfu Wang's group seminar
Beijing International Center for Mathematical Research, Peking University, Beijing, China

Interacting Particle Systems and Their Applications *Oct 5–16, 2026*
Invited talk: "Uniform-in-time propagation of chaos and long-time asymptotics for CBO"
Institute for Mathematical Sciences, National University of Singapore, Singapore

Partial Differential Equations for Many Particles Systems *Aug 2026*
Poster: "Uniform-in-time Propagation of Chaos for First- and Second-Order CBO"
ICERM, Brown University, Providence, RI, USA

BIRS–IASM Workshop: Interacting Particle Systems: Theoretical Innovations and Practical Applications *Jun–Jul 2026*
Invited talk (virtual): "Uniform-in-time Propagation of Chaos for First- and Second-Order CBO"
Institute for Advanced Study in Mathematics, Hangzhou, China

13th Summer School on Methods and Models of Kinetic Theory (M&MKT 2026) *Jun 2026*
Poster: "Uniform-in-time Propagation of Chaos for First- and Second-Order CBO"
Pesaro, Italy

SIAM Conference on Uncertainty Quantification (UQ26) *Mar 2026*
Invited talk: "Uniform-in-time Propagation of Chaos for Consensus-Based Optimization"
Minneapolis, MN, USA

Pathways Workshop & Introductory Workshop: Kinetic Theory & Stochastic Partial Differential Equations *Aug 2025*
Participant, Simons Laufer Mathematical Sciences Institute, Berkeley, CA, USA

Frontiers in Applied Analysis *Jun 2025*
Poster: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Carnegie Mellon University, Pittsburgh, PA, USA

SIAM Conference on Applications of Dynamical Systems *May 2025*
Minisymposium co-organizer and speaker
Co-organized with Nicolás García Trillos (UW–Madison), Sixu Li (UW–Madison), Claudia Totzeck (University of Wuppertal), and Yuhua Zhu (UCLA)
Talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Minisymposia MS 47, 65, 83: "Collective Dynamics in Multi-Agent Systems: Advances in Learning and Optimization"
Denver, CO, USA

AIM Workshop: Integro-differential Equations in Many-Particle Interacting Systems *Apr 2025*
Participant, American Institute of Mathematics, Pasadena, CA, USA

SOCAMS 2025 *Apr 2025*
Invited talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
University of California, Riverside, CA, USA

SIAM Conference on Mathematics of Data Science *Oct 2024*
Poster: "Recent Developments in Consensus Based Optimization (CBO)"
Minisymposium MS 27: "Interacting Particle Systems in Data Science: From Theory to Applications"
Atlanta, GA, USA

Junior Kinetic Theory Workshop *Sep 2024*
Invited talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Yonsei University, South Korea

9th European Congress of Mathematics (ECM) *Jul 2024*
Invited talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Seville, Spain

Modern Perspectives in Applied Mathematics: Theory and Numerics of PDEs *Jul 2024*
Participant, ETH Zurich, Switzerland

CIRM: Mathematical Biology: Collective Behavior and Pattern Formation *Jul 2024*
Poster: "Mono-cluster flocking and uniform-in-time stability of the discrete Motsch–Tadmor model"
CIRM, Marseille, France

CIRM: Frontiers in Interacting Particle Systems, Aggregation-Diffusion Equations & Collective Behavior *Jun 2024*
Participant, CIRM, Marseille, France

SOCAMS 2024 *Apr 2024*
Talk: "Recent developments in Consensus-Based Optimization (CBO)"
University of California, San Diego, CA, USA

International Congress on Industrial and Applied Mathematics (ICIAM) *Aug 2023*
Participant, Waseda University, Tokyo, Japan

Purpose-Driven Particle Systems *Mar 2023*
Participant, Lorentz Center, Leiden, Netherlands

HONORS & AWARDS

Graduation with Highest Distinction
Graduation ceremony valedictorian — Yonsei University, South Korea

Amore Pacific President Award
Alumni Association Presidential Award — Yonsei University, South Korea

Silver Prize
National Math Olympiad for Undergraduate Students, Korean Mathematical Society, South Korea
Non-mathematics-major division for engineering, medical, and other science students

Awards of Highest Academic Honor and Academic Honor
Top 1% and top 3% of students in the department, respectively
Yonsei University, South Korea

FELLOWSHIPS & GRANTS

Oberwolfach Travel Grant *Dec 2026*
Support for the workshop "Variational Methods for Evolution"
Mathematisches Forschungsinstitut Oberwolfach, Germany

BICMR Research Visit Lodging Support *Oct 2026*
Host-provided lodging support for a research visit with Prof. Zhenfu Wang
Beijing International Center for Mathematical Research, Peking University, China

IMS Lodging Support *Oct 2026*
Support for "Interacting Particle Systems and Their Applications"
Institute for Mathematical Sciences, National University of Singapore, Singapore

ICERM Lodging Support *Aug 2026*
Support for "Partial Differential Equations for Many Particles Systems"
ICERM, Brown University, USA

M&MKT Summer School Lodging Support *Jun 2026*
Support for the 13th Summer School on Methods and Models of Kinetic Theory
M&MKT, Pesaro, Italy

Caltech Y Hummel Gray Award *Nov 2025*
Travel award for graduate students — Caltech, USA

Travel Support for SLMath Workshops *Aug 2025*
Support to attend the Pathways Workshop and Introductory Workshop on Kinetic Theory & Stochastic Partial Differential Equations
Simons Laufer Mathematical Sciences Institute, Berkeley, CA, USA (workshop venue)

CMU Travel Support *Jun 2025*
Support for "Frontiers in Applied Analysis" — Carnegie Mellon University, USA

SIAM Travel Award *May 2025*
Support for the Conference on Applications of Dynamical Systems — SIAM, USA

ETH Zurich Conference Travel Support *Jul 2024*
Support for "Modern Perspectives in Applied Mathematics: Theory and Numerics of PDEs"
ETH Zurich, Switzerland

CIRM Conference Travel Support *Jun–Jul 2024*
Travel and lodging support for "Frontiers in Interacting Particle Systems" and "Mathematical Biology: Collective Behavior and Pattern Formation"
CIRM, Marseille, France

Two-Year Departmental Fellowship *Sep 2021 – Sep 2023*
Computing and Mathematical Sciences departmental fellowship — Caltech, USA

OSE Tutoring Scholarship
Award for successful tutoring — OSE Center, Yonsei University, South Korea

Merit-Based Undergraduate/Graduate Scholarship
External scholarship — Kwanjeong Foundation, South Korea

Merit-Based Undergraduate Scholarship
External scholarship
Korean Foundation for Advanced Studies (KFAS), SK, South Korea

Academic Scholarship
University and private-fund scholarships — Yonsei University, South Korea

Academic Scholarship (declined)
Awarded to admitted undergraduates with exceptional academic achievements — Korea University Business Department, South Korea

TEACHING EXPERIENCE

All teaching assistant appointments below were at the California Institute of Technology, USA.

TA, ACM 270: Particles in Motion—Mathematical Modelling Across Scales *Apr–Jun 2026*
Instructor: Prof. Kathrin Hellmuth; graduate-level course
Developed lecture notes and delivered two 1.5-hour lectures on mean-field limits and propagation of chaos; held weekly office hours; developed and graded homework problems.

TA, ACM 95/100A: Introductory Methods of Applied Mathematics for the Physical Sciences *Jan–Mar 2026*
Instructor: Prof. Konstantin Zuev; undergraduate/graduate-level course
Held weekly office hours; graded weekly homework problems and exams.

TA, ACM 256: Optimal Transport *Jan–Mar 2025*
Instructors: Prof. Franca Hoffmann and Prof. Ricardo Baptista; graduate-level course
Held weekly office hours; graded homework problems.

TA, ACM 95/100A: Introductory Methods of Applied Mathematics for the Physical Sciences *Jan–Mar 2024*
Instructor: Prof. Franca Hoffmann; undergraduate/graduate-level course
Held weekly office hours; graded weekly homework problems and exams.

TA, CMS/ACM 117: Probability Theory and Stochastic Processes *Sep–Dec 2023*
Instructor: Prof. Joel Tropp; graduate-level course
Graded weekly homework problems and exams.

TA, ACM 105: Functional Analysis *Jan–Mar 2023*
Instructor: Prof. Oscar Leong; graduate-level course
Delivered two 1.5-hour lectures on the Hahn–Banach theorem and the open mapping and closed graph theorems; held weekly office hours; graded homework problems and exams.

TA, CMS/ACM 117: Probability Theory and Stochastic Processes *Sep–Dec 2022*
Instructor: Prof. Joel Tropp; graduate-level course
Held weekly office hours; wrote problem-set solutions; graded weekly homework problems and exams.

Private Tutoring
Tutored single- and multivariable calculus for a first-year undergraduate — Seoul, South Korea

Peer Tutoring
Tutored undergraduate peers in Analysis 1 — OSE Center, Yonsei University, South Korea

MENTORSHIP

Ritvik Teegavarapu, Caltech SURF mentee, USA *May 2024 – present*

Co-mentored with Prof. Franca Hoffmann on particle swarm optimization; continuing collaboration on annealed consensus-based optimization and sampling

Student awards: SOCAMS Best Poster Award (2025); Caltech's George W. and Bernice E. Green Memorial Prize (research award, 2026)

Now a Ph.D. student with Prof. Kavita Ramanan at Brown University (since fall 2026)

LEADERSHIP & ACADEMIC SERVICE

SIAM Student Chapter, Caltech, USA *2025 – present*

President (2025–26); committee member (2026–27)

Womxn in Engineering and Applied Science, Caltech, USA *2024 – present*

Committee member; organized kick-off events, faculty lunches and coffee conversations, and social gatherings.

Research Group Meeting Organizer, Caltech, USA *Spring 2025*

Organized meetings for Prof. Franca Hoffmann's research group.

Head Proctor, First-Year Preliminary Exam, Caltech, USA *Winter 2023*

Coordinated mock exams, matched students with proctors, and collected and shared past exams.

OUTREACH & SCIENCE COMMUNICATION

Three Minute Thesis (3MT) Competition, Caltech, USA *April 2026*

Finalist; the only finalist from Mathematics or Applied and Computational Mathematics in 2026.
Presented consensus-based optimization research to a general audience.

AIM Math Fair, American Institute of Mathematics, USA *June 2025*

Volunteer; guided visiting families, from young children to adults, through mathematical puzzles.

PROGRAMMING & LANGUAGES

Programming and typesetting: Python, MATLAB, LaTeX
Languages: Korean (native); English (fluent); French (intermediate; DELF B1, B2)